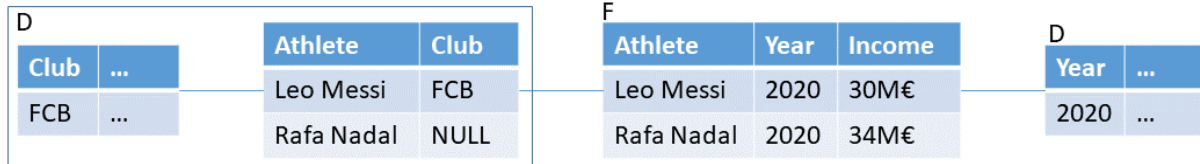


Family name: _____ Given name: _____

20%

1. Identify the main problem in the multidimensional schema below, and propose a solution to solve it. You should assume athletes do not change the club in the middle of the year.



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2. Explain the difference between “Time-variant” and “Nonvolatile” characteristics in the DW definition of W. Inmon. Illustrate it with an example different from that in the materials of the course.

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3. Which is the worst-case cost of the R-Swoosh algorithm and when would it happen?

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4. Give an example of schematic discrepancy (different from that in the slides), where data in one source is represented as metadata in the other. Draw and briefly explain it.

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5. Consider a table of 1000 bytes and a block size of 250 bytes. Given the following bitmap index over attribute A using run-length encoding with Hakan factor three, identify which rows are in the result of a selection over the corresponding table with condition $A='y'$. For each row, you should simply indicate its block and position inside the block.

x	y	z	t
0	0	1	1
1	0	1	0
1	1	0	1
0	1	1	0
1	0	0	0
1	1	0	0
1	1	0	1
1	0	1	1
0	0	1	0
1	1	1	1
0		0	0
0		1	0
		0	
		0	